



Project M5: Thermocouple Response Analysis

- Team 4
- ENGR 1560
- Final Project Presentation

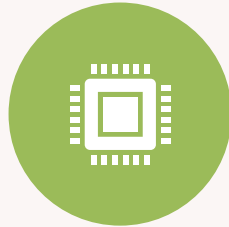
What the Project Is About

- • Study 5 thermocouple models
 - • Measure how fast each reacts to temperature changes
 - • Compare performance to price
 - • Improve our algorithm from earlier projects
- 

What FOS Wants to Know



- HOW CONSISTENT ARE THE SENSORS?



- DO FASTER SENSORS COST MORE?



- IS OUR ALGORITHM ACCURATE?



- CAN WE TRUST THE DATA?

What a Thermocouple Does



- MEASURES TEMPERATURE



- HAS A SPEED CALLED THE 'TIME CONSTANT' (T)



- SMALLER T = FASTER RESPONSE

How Our Algorithm Works

1

1. Smooth the data

2

2. Detect heating or cooling

3

3. Find starting and ending temperatures

4

4. Find where the temperature starts changing

5

5. Fit an exponential curve to estimate τ

Key Improvements We Added



- BETTER T ESTIMATION
USING CURVE FITTING



- HANDLES NOISY DATA
SMOOTHER



- FASTER ALGORITHM
(0.0079 S PER RUN)

Sample Results: Calibration

- Heating Clean $\tau = 0.34$ s
- Cooling Clean $\tau = 1.78$ s
- Heating Noisy $\tau = 3.22$ s
- Cooling Noisy $\tau = 2.70$ s

What We Learned From Calibration

- Clean data gives reasonable τ values

- Noisy data makes τ much larger

- Early noise confuses the algorithm

100 Dataset Results

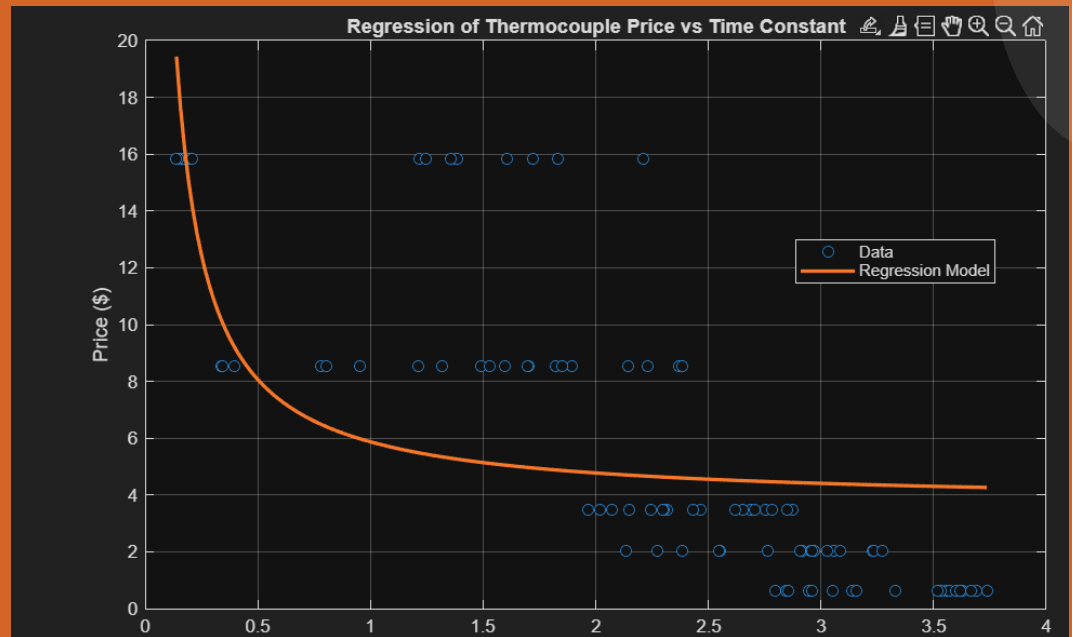
FOS-1: Fastest ($\tau \approx 0.80$ s)

FOS-5: Slowest ($\tau \approx 3.31$ s)

FOS-3 & FOS-4 very consistent

Regression Results

- Price vs $1/\tau$
- $r^2 = 0.4536$



Error in the Process



- Noise strongly affects τ



- Early-time data is the biggest issue

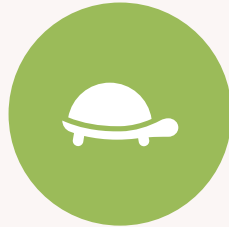


- Even clean data doesn't give perfect fits

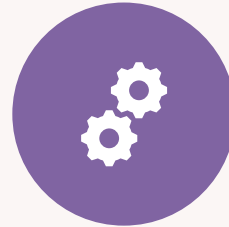
What FOS Can Honestly Claim



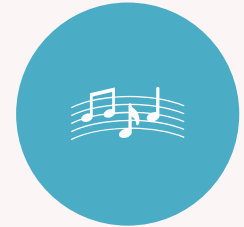
- FASTER SENSORS
REALLY DO COST
MORE



- HIGHER MODELS
RESPOND SLOWER



- MANUFACTURING
IS CONSISTENT



- PERFORMANCE
DIFFERENCES ARE
REAL

Why Our Algorithm Works for FOS



- FAST PROCESSING



- HANDLES ALL 100 DATASETS



- GIVES TRENDS THAT MATCH REAL EXPECTATIONS

Conclusions



- τ increases from FOS-1 to FOS-5



- Noise affects measurements



- Our algorithm is reliable and fast



- Price is related to performance